

GPT Caddy

A personal project to make GPT Golf easier.
See inside the run. Decide what to change next.

I built it for myself — every GPT Golf run goes through it.

LIVE DEMO

REAL RUN

~18 MIN + Q&A



NEED

Every GPT Golf run is guess-and-check.

Tweak → **wait** → **inspect loss** → **guess again**

A LOSS CURVE



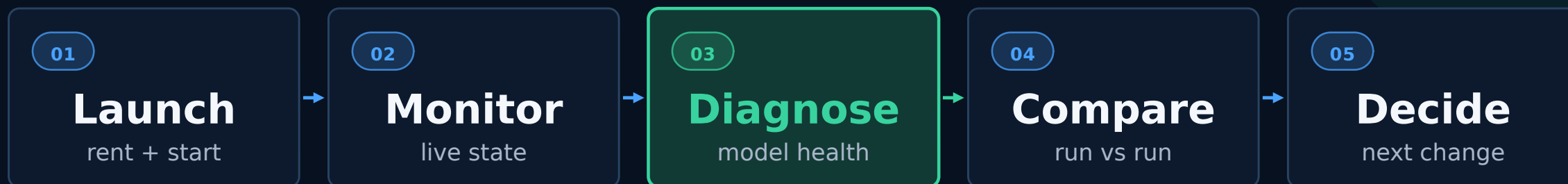
tells me what happened

— **not
why.**

So I built the instrument I wanted.

See why. Decide in minutes.

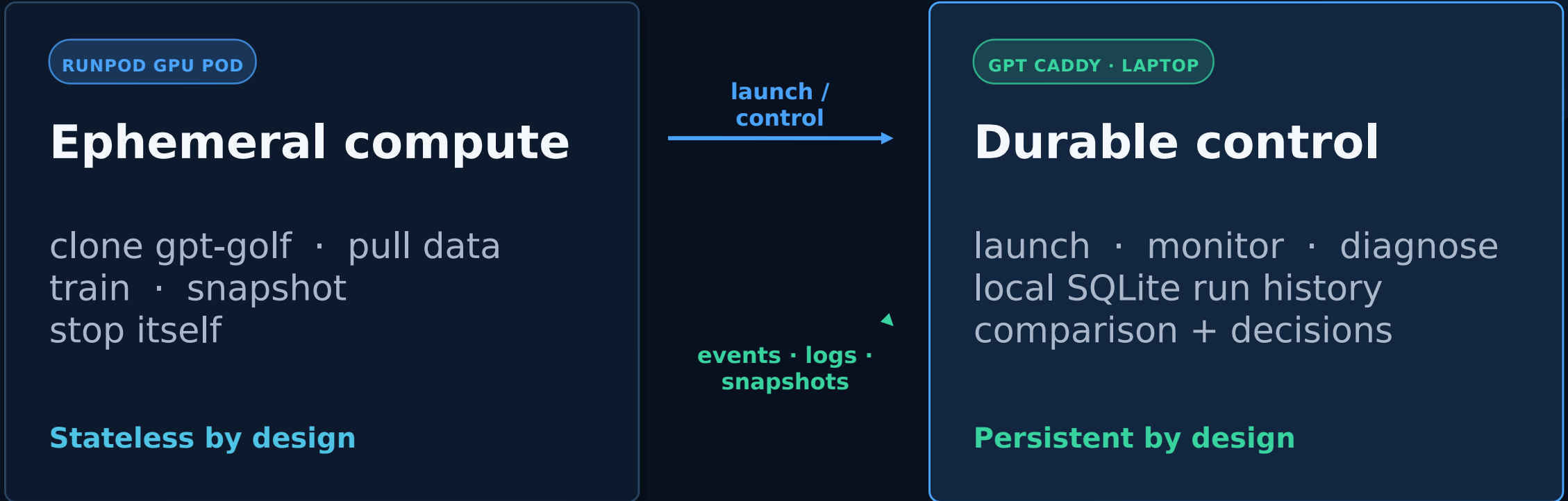
This loop is the whole tool.



Every pass through the loop should reduce uncertainty.

Slides for the how. The live app for the proof. **A pod is training right now.**

Ephemeral GPU. Durable control plane.



The pod can disappear. The durable run history survives.

The build gets faster by removing the wrong work.

C++ SENTENCEPIECE
FORK

6x+ faster tokenization

10B tokens **>1 hour → <10 minutes**

~50% less memory

Parallel buffering + divide-and-conquer over the corpus.

CUDA / LOADER

700+ MB removed from the image

16 KB stubs satisfy the loader

~500 MB FFT / sparse + ~206 MB NCCL

Derived from the installed Torch binary. Multi-GPU restore fails loudly.

Bake Torch / CUDA once. Rebuild application code in seconds.

A small event path keeps the live view honest.



- **Boots before dependencies**

stdlib-only daemon

- **Every event is durable**

persisted to SQLite

- **One stream, three uses**

browser · history · diagnostics

Pod → authenticated JSONL → laptop pipeline → browser / SQLite / diagnostics

LIVE

Watch three things move.

01

Race vs baseline

Ahead or behind at the same budget?

02

Cost in real time

Meter ticks; the pod stops itself.

03

Model health

Metrics live; the verdict runs off-GPU.



A real run, updating now. The slide is only the map.

DIAGNOSE

The diagnostics check their own arithmetic.

01

RECOMPUTE

layer outputs

02

RE-DERIVE

loss independently

03

CROSS-CHECK

BPB vs on-pod eval

gg-main-0622-205629 · main step 1000 just now

WARN

Deep Dive

WARN

3 checks need review in capacity and flow.

4 healthy 3 warnings 0 failing 0 unavailable 2 info

Capacity

1 good · 2 warn

Dead neurons 7.2% Head redundancy 0.636

Attn entropy (min) 1.528

Flow

1 good · 1 warn · 1 info

Mean loss 2.778 Residual growth 0.375

Logit saturation 0.0%

Gradients

1 good

Grad norm ratio 11.436

Architecture

1 good · 1 info

Embed eff_rank 212.519 Zero-init escape (min) 4.281

1000

Latest

NEEDS ATTENTION

Dead neurons

7.2%

Units that never activate waste capacity; a large fraction means dead width. · warn ≥ 0.05 , bad ≥ 0.2

Head redundancy

0.636

Attention heads computing near-identical outputs are redundant capacity. · warn ≥ 0.6 , bad ≥ 0.85

Residual growth

0.375

Residual stream norm should grow moderately; too flat or exploding is unhealthy. · ok 0.5–4

GPU trains. Laptop audits. Anything unverified is flagged.

COMPARE

Did my change help?



01

Did loss improve?

curves + no-regression banner

02

Did validation improve?

BPB at equal progress

03

Did model health improve?

verdict + headline metrics

Compare at equal step or equal progress. Then decide what to change next.

Launch → Monitor → Diagnose → **Compare** → Decide ↻

CLOSE

From an unmet need to a tool I use every day.

**I found the need, built the tool,
and made it trustworthy enough to run on
every GPT Golf experiment.**

need → tool → evidence → decision

The GPU stops itself. The run history and diagnosis remain.



Appendix / backup

Static live-demo fallbacks · rigor · deeper technical discussion

PULL UP ON DEMAND

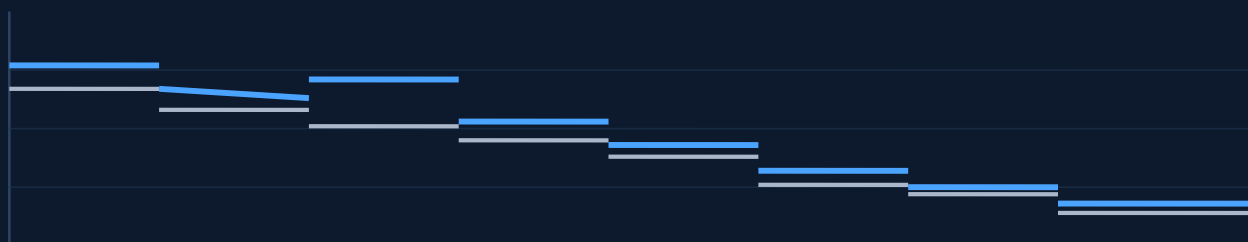
Static fallback: the run at a glance.

GPT Caddy / training run

RUNNING

Loss vs baseline

live run baseline



Race
ahead / behind

Cost
ticks live

ETA
derived

Health
pending verdict

live events / logs

Starting training
step events → metrics → snapshot → diagnostics

Static fallback: the verdict and its checks.

MODEL HEALTH



**CROSS-
CHECKED**

No result is silently
trusted.

GPU trains
Laptop audits



Recompute

layer outputs from components



Re-derive

loss independently



**Cross-
check**

BPB vs on-pod evaluation

Thresholds come from one backend metric registry.

Static fallback: did the change help?

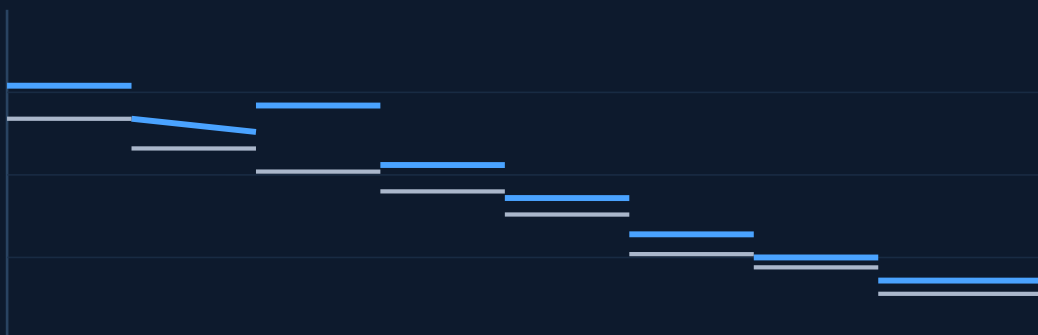
RUN A

RUN B

EQUAL PROGRESS

Loss overlay

Run B Run A



LOSS

lower / higher

HEALTH

verdict + headlines

INTERNALS

which blocks moved

Decide what to change next.

Built to be trusted.

01

Build integrity

Torch-derived stubs
Fail-loud restore

02

Data-path integrity

Schema contract
Persistence + redaction

03

Diagnostic integrity

Self-checking math
BPB cross-check

Automated checks

pytest: health verdicts · compare diffs · schema · golden end-to-end diagnostics · DB / pipeline / routes · secret redaction

frontend node --test: derived metrics + diagnostics logic · CI: Python, frontend, and image tests on every push

Next hardening: static typing · coverage floor · failure injection at thread / SQLite / subprocess seams · dependency scanning

Technical proof: github.com/sysreq/sentencepiece_parallel